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QUESTIONS 10 / 10 completed

Linear Search

LINEAR SEARCH

Create a C program to perform Linear Search on an array of 10 integers. The program should allow manual input or random generation of array elements. It must display the first position of the element (1-based indexing), total

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, target, pos = -1, comparisons = 0;
5
6     printf("Enter size: ");
7     scanf("%d", &n);
8
9     int arr[n];
10
11    printf("Enter array elements: ");
12    for(i = 0; i < n; i++)
13        scanf("%d", &arr[i]);
14
15    printf("Enter target: ");
16    scanf("%d", &target);
17
18    printf("Comparison Trace:\n");
19    for(i = 0; i < n; i++) {
20        comparisons++;
21        printf("%d %s %d\n", arr[i], (arr[i] == target) ? "=" : "!=", target);
```

INPUT

5
10 20 30 40 50
40

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QUESTIONS 10 / 10 completed

Binary Search

BINARY SEARCH

"Write a C program implementing binary search on a sorted array of 15 integers. The program must: (1) accept an unsorted array and sort it first, (2) perform binary search, (3) display search iterations with mid-point calculations, (4)

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[10];
5     int i, target;
6     int low = 0, high = 9, mid;
7     int comparisons = 0;
8     int found = -1;
9
10    printf("Enter 10 sorted elements:\n");
11    for(i = 0; i < 10; i++)
12        scanf("%d", &arr[i]);
13
14    printf("Enter target: ");
15    scanf("%d", &target);
16
17    while(low <= high) {
18        mid = (low + high) / 2;
19        comparisons++;
20
21        if(arr[mid] == target) {
```

INPUT

10
5 10 15 20 25 30 35
40 45 50
35

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QUESTIONS 10 / 10 completed

Binary Search- Iterative

BINARY SEARCH-ITERATIVE

Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day >=30°C and last day <=25°C. Display temperatures, indices (0-based), or ""Not found"". Validate sorted; n<=50.

PROGRAM EDITOR

C Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[15], n = 15;
5     int i, j, temp, target;
6
7     printf("Enter 15 elements:\n");
8     for(i = 0; i < n; i++)
9         scanf("%d", &arr[i]);
10
11     for(i = 0; i < n - 1; i++) {
12         for(j = 0; j < n - i - 1; j++) {
13             if(arr[j] > arr[j + 1]) {
14                 temp = arr[j];
15                 arr[j] = arr[j + 1];
16                 arr[j + 1] = temp;
17             }
18         }
19     }
20
21     printf("Sorted Array:\n");
```

INPUT

40 10 60 20 80 70 30
50 90 100 5 15 35 45
55
35

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Search

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QUESTIONS 10 / 10 completed

Linear Search -Sort

LINEAR SEARCH -SORT

Read n book page counts into an array. Sort ascending using any method. Use linear search to find the first book having >= 300 pages and the last book having <= 150 pages. Display page counts and indices (0-based), or ""Not found"". Validate

PROGRAM EDITOR

C Run Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i;
5     float temp[50];
6
7     printf("Enter n: ");
8     scanf("%d", &n);
9
10    if(n > 50) {
11        printf("Invalid Input\n");
12        return 0;
13    }
14
15    printf("Enter temperatures:\n");
16    for(i = 0; i < n; i++)
17        scanf("%f", &temp[i]);
18
19    int index = -1;
20
21    for(i = n - 1; i >= 0; i--) {
```

INPUT

7
22.5 19.0 31.5 18.5
25.0 17.0 29.5

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Search

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QUESTIONS 10 / 10 completed

Linear Search - Anagrams

LINEAR SEARCH - ANAGRAMS

"Read n strings, use linear search to find all anagrams of a target string (same chars, any order). Display positions (1-based) and matching strings. Case-insensitive; n <= 50. Sample Input: 5
1) listen silent
2) race care"

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 void sortString(char s[]) {
6     int i, j, len;
7     char temp;
8
9     len = strlen(s);
10
11    for(i = 0; i < len; i++)
12        s[i] = tolower(s[i]);
13
14    for(i = 0; i < len - 1; i++) {
15        for(j = i + 1; j < len; j++) {
16            if(s[i] > s[j]) {
17                temp = s[i];
18                s[i] = s[j];
19                s[j] = temp;
20            }
21        }
22    }
```

INPUT

5
listen
silent
race
care
hello
listen

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QUESTIONS 10 / 10 completed

Linear Search on A

LINEAR SEARCH ON ARRAY

"Develop a C program to perform Linear Search on an array of N integers. Accept array size dynamically. Sample Input/Output
Enter size: 5
Array: 10 20 30 40 50
Enter target: 40
Comparison Trace:
10 != 40"

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, i, key, found = 0;
6
7     printf("Enter size: ");
8     scanf("%d", &n);
9
10    int arr[n];
11
12    printf("Enter array elements: ");
13    for(i = 0; i < n; i++)
14        scanf("%d", &arr[i]);
15
16    printf("Enter target: ");
17    scanf("%d", &key);
18
19    printf("Comparison Trace:\n");
20
21    for(i = 0; i < n; i++)
```

INPUT

Enter size: 5
Enter array elements:
10 20 30 40 50
Enter target: 40

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QUESTIONS 10 / 10 completed

Binary Search - Arr

BINARY SEARCH - ARRAY

Write a C program to perform Binary Search on a sorted array of 10 integers. The program should accept the array elements and a target value from the user. It must display the position of the element and the total number of comparisons made.

PROGRAM EDITOR

C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int a[10];
6     int i, key;
7     int low = 0, high = 9, mid;
8     int comparisons = 0;
9
10    printf("Enter 10 sorted integers:\n");
11    for(i = 0; i < 10; i++)
12        scanf("%d", &a[i]);
13
14    printf("Enter element to search: ");
15    scanf("%d", &key);
16
17    while(low <= high)
18    {
19        comparisons++;
20        mid = (low + high) / 2;
21    }
```

INPUT

Enter 10 sorted integers:
5 12 18 23 34 45 56
67 78 90
Enter element to search:
45

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Search

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QUESTIONS 10 / 10 completed

Linear Search - Tem

LINEAR SEARCH - TEMPERATURE

"Read n daily temperatures into an array. Use Linear Search to find the last day whose temperature is less than 20°C. Display temperatures, index (0-based), and value. If not found, print "Not found". Validate n ≤ 50.

PROGRAM EDITOR

C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, temp[50];
6     int i, index = -1;
7
8     printf("Enter number of days: ");
9     scanf("%d", &n);
10
11    if(n < 1 || n > 50)
12    {
13        printf("Invalid input");
14        return 0;
15    }
16
17    printf("Enter temperatures:\n");
18    for(i = 0; i < n; i++)
19        scanf("%d", &temp[i]);
20
21    printf("Temperatures: ");
```

INPUT

Enter number of days:
7
18 25 21 19 30 17 24

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Search

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QUESTIONS 10 / 10 completed

Linear Search - Prices

Read n product prices into an array. Sort ascending using any method. Use linear search to find the first price $\geq ₹500$ and the last price $\leq ₹200$. Display prices and indices (0-based), or "Not found". Validate sorted order; $n \leq 50$.

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, price[50];
6     int i, j, temp;
7     int first500 = -1, last200 = -1;
8
9     scanf("%d", &n);
10
11     if(n < 1 || n > 50)
12     {
13         printf("Invalid input");
14         return 0;
15     }
16
17     for(i = 0; i < n; i++)
18         scanf("%d", &price[i]);
19
20     for(i = 0; i < n - 1; i++)
21     {
```

INPUT

Enter number of products: 7
150 650 220 90 500
780 180

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QUESTIONS 10 / 10 completed

Bin. Search-Iterativ

BIN. SEARCH-ITERATIVE-TEMP

Read n daily temperatures into an array, sort ascending using any method. Use binary search (iterative) to find first day $\geq 30^\circ\text{C}$ and last day $\leq 25^\circ\text{C}$. Display temperatures, indices (0-based), or "Not found". Validate sorted; $n \leq 50$.

PROGRAM EDITOR C Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n, a[50];
6     int i, j, temp;
7
8     printf("Enter number of days: ");
9     scanf("%d", &n);
10
11     if(n < 1 || n > 50)
12     {
13         printf("Invalid input");
14         return 0;
15     }
16
17     printf("Enter temperatures:\n");
18     for(i = 0; i < n; i++)
19         scanf("%d", &a[i]);
20
21     // Bubble Sort
```

INPUT

6
22 28 31 24 29 32

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